

Rohan Kumar

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EDUCATION

University of Illinois at Urbana-Champaign

Expected Graduation Date: December 2026

Bachelor of Science in Computer Science and Statistics (GPA: 3.86 / 4.00)

Urbana-Champaign Area, IL

Courses: Object Oriented Programming, Data Structures, Algorithms, Computer Systems, Database Systems, Distributed Systems, High Frequency Trading Technology, Statistical Modeling, Statistical Learning, Applied Machine Learning

SKILLS

Programming Languages: Python, TypeScript, JavaScript, Java, C/C++, C#, SQL, R, HTML/CSS

Frameworks & Libraries: React, Node.js, .NET, FastAPI, SQLAlchemy, Playwright, Flask, TensorFlow, Scikit-Learn, Prisma

Cloud & DevOps: AWS (EC2, Glue), GCP, Fly.io, Cloudflare, Azure, Docker, Git, GitHub Actions, Spark, CI/CD

Tools & Concepts: LLMs, LLM Evaluation, pgvector, Observability, Multi-Agent Systems, Retrieval-Augmented Generation (RAG), Vector Databases, Distributed Systems, RESTful APIs, Test-Driven Development, Agile, IDOR, Threat Modeling

EXPERIENCE

Founding Engineer

August 2026 – Present

Vinskal

- Architected the continuous job-discovery pipeline behind 130K+ live job postings using PostgreSQL lease claims and per-host rate pacing; diagnosed a 50-hour, 49K+ row silent stall and redesigned the queue for durable recovery.
- Built an agentic incident responder (Cloudflare Workers, GitHub Actions) and its production alert-routing middleware, autonomously investigating downtime, quota, and runtime failures and opening PRs for human review (1K+ alerts handled).
- Built a Playwright-based browser job application agent supporting five ATS platforms and validated on live employer submissions; developed its eval harness with a seed/held-out split, model judges, scorecards, and per-run budget ceilings.
- Cut monthly infrastructure spend 24% by eliminating \$281/mo in runaway database egress, migrating CI to merge-queued UbiCloud runner architecture that also ran 40% faster, and adding model/vendor usage metering.
- Technologies:** Python, TypeScript, FastAPI, PostgreSQL, Playwright, Docker, Cloudflare Workers, GitHub Actions

Software Engineer Intern

May 2026 – August 2026

Pinterest

New York, NY

- Built a multi-agent AI system (TypeScript, LangGraph) that autonomously searches Pinterest's source code for security vulnerabilities, grounded in a retrieval (RAG) layer over the company's own history of confirmed vulnerabilities—surfacing a Critical-severity vulnerability that triggered a security incident and ≈ 12 low-medium severity bugs as of August 2026.
- Engineered a second agentic harness that dynamically validates suspected vulnerabilities by driving a Chrome browser, logging into real accounts, attempting to perform the claimed attack, and returning a Reproduced/Refuted verdict.
- Owned the platform E2E as the sole engineer building the infrastructure beneath both systems: a fleet-wide rate governor and a Postgres-backed job queue that keep hours-long agent runs alive and durable under a shared LLM token budget.
- Technologies:** TypeScript, Python, PostgreSQL, Prisma, LangGraph, DeepAgents, RAG, LLMs, Docker, REST APIs, Git

Software Development Engineer Intern

February 2026 – April 2026

Amazon

Seattle, WA

- Designed and deployed multilingual personalization features for Amazon Music's voice recommendation system (≈ 23 M+ daily requests), integrating user listening behavior into an ML ranking pipeline and achieving 96%+ feature coverage.
- Owned end-to-end system design and development of a language-aware candidate filtering system in Java, reducing irrelevant cross-language recommendations and improving music recommendation quality across 17M+ daily voice requests.
- Engineered 5 large-scale PySpark data pipelines (AWS Glue) to analyze 100M+ recommendation events, uncovering feature coverage gaps and critical quality issues that directly informed ranking model inputs and system design decisions.
- Technologies:** Java, PySpark, AWS (Glue), Reinforcement Learning (RL), A/B Testing, Distributed Systems

PROJECTS

Market Data Feeds Compression (PCAP Compression) – FinTech Lab @ UIUC | Technical Lead

- Designed and implemented a high-performance C++17 market data system for IEX TOPS/DEEP and NASDAQ ITCH PCAP feeds, using Zstandard (Zstd) with dictionary training to reduce data size by 80-85% ($\approx 15\%$ gain over generic Zstd).
- Engineered a streaming, memory-efficient architecture (≤ 256 MB) capable of processing multi-GB captures at 40K+ packets/sec, suitable for continuous market-hours ingestion while maintaining fast decompression and system reliability.
- Technologies:** C++17, libpcap, Zstandard (Zstd), Make, Linux, Data Infrastructure, High Frequency Trading (HFT)

LEADERSHIP

Technology Director

May 2025 – December 2025

National Organization for Business and Engineering (NOBE): Illinois Chapter

Urbana-Champaign Area, IL

- Led a team of 40+ student software engineers across 4+ technical projects for a diverse portfolio of clients.